

Health & Education

Readings

The impact of artificial intelligence on learner–instructor interaction in online learning

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26 October 2021

Artificial Intelligence in Healthcare pp. 25 – 60

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2020

Paper: The impact of artificial intelligence on learner–instructor interaction in online learning**Purpose of the study**

- the **researchers** wanted to **understand**
 - how **AI** systems **change** interactions between **students** and **instructors** in online **learning**
 - how **students** and **instructors** perceive these **changes**
 - what **opportunities** and **risks** **AI** introduces in **educational** settings
- **AI systems** are increasingly used to
 - **personalize** learning
 - **automate grading** and **feedback**
 - support **communication** in online courses
- however, their effect on **human interaction in learning environments** is not yet fully understood

Key Concepts

- **instructional interaction**
 - explaining **concepts**
 - providing **feedback**
 - guiding **learning** activities
- **social interaction**
 - building **relationships**
 - encouraging **participation**
 - maintaining **engagement**
- **these interactions strongly affect**
 - student **motivation**
 - **satisfaction** with courses
 - learning **outcomes**
- **AI technologies may change how these interactions occur**

Research Method	- the study used a design method called “speed dating” - this method allows participants to quickly experience and evaluate different AI concepts - participants interacted with multiple hypothetical AI systems designed for online education. - examples included systems that could monitor student progress recommend learning materials - automatically answer questions provide personalized feedback - researchers collected participant reactions and perceptions
Types of AI Roles in Education	- AI as a teaching assistant - AI helps instructors by grading assignments answering routine questions - providing automated feedback - this reduces instructor workload - AI as a learning companion - AI directly supports students by recommending resources explaining difficult concepts guiding practice exercises - this provides personalized learning support - AI as an administrative tool - AI may handle logistical tasks such as tracking participation monitoring performance identifying struggling students - this helps instructors manage large online classes.
Key Findings	- AI can improve efficiency - participants believed AI could automate repetitive tasks speed up feedback

	▪ help instructors manage large courses • this may allow instructors to focus on higher-level teaching tasks • AI can personalize learning ○ AI systems can ▪ adapt content to individual learners ▪ provide tailored feedback ▪ recommend resources based on performance • personalization may improve student engagement and learning outcomes • human interaction Is still important ○ participants emphasized that ▪ students still need human instructors ▪ emotional support and mentorship cannot easily be replaced • AI should support instructors, not replace them. • participants expressed concerns about ○ reduced human contact ▪ too much automation may reduce meaningful interaction ○ privacy and data use ▪ AI systems require large amounts of student data ○ over-reliance on technology ▪ institutions might rely too heavily on automated systems
Implications for Education	• AI should augment teaching, not replace instructors ○ best approach ○ human + AI collaboration ○ example model ▪ instructor focuses on • mentoring • complex feedback • social interaction ▪ AI handles

	• routine tasks • data analysis • personalized recommendations
Future Research Needs	• artificial intelligence has the potential to transform online education • its success depends on maintaining meaningful human interaction between learners and instructors • AI works best as a support tool that enhances teaching rather than replacing it
Chapter: Artificial Intelligence in Healthcare	
Main Goal of the Chapter	• how artificial intelligence is used in education • what AI systems can do for teaching and learning • the benefits, risks, and future directions of AI-based learning systems • AI is becoming a major component of digital learning environments, especially in online and large-scale education systems
Artificial Intelligence in Education (AIED)	• AEID refers to systems that can ○ analyze student data ○ adapt learning materials ○ support instructors ○ automate educational tasks • examples include ○ intelligent tutoring systems ○ automated grading tools ○ recommendation systems for learning content ○ conversational agents or chatbots • these systems attempt to replicate or support some teaching functions

**Key Capabilities
of AI in Learning
Systems**

- **adaptive** learning
 - **analyze** student **performance**
 - **detect** knowledge **gaps**
 - **adjust** **difficulty** levels
- each student **receives** personalized **instruction**
- **intelligent tutoring**
 - **guide** students **step-by-step**
 - **give** **immediate** feedback
 - **monitor** learning **progress**
- this mimics **one-on-one tutoring**
- automated **assessment**
 - **grade** assignments
 - evaluate **short answers** or **essays**
 - **provide** **feedback** quickly
- this **reduces** instructor **workload**
- **learning analytics**
 - AI systems **analyze large** educational **datasets**
 - **identify** **struggling** students
 - **detect** learning **patterns**
 - **improve** **course design**
- these insights **help instructors** intervene **earlier**

Benefits of AI in Education	• personalization ○ customized pacing ○ adaptive difficulty ○ targeted feedback • learn according to individual needs • scalability ○ thousands of students simultaneously ○ large online courses (MOOCs) • enables mass education with individualized support • continuous feedback ○ provide feedback immediately ○ provide guidance throughout the learning process • immediate feedback improves learning efficiency • instructor support ○ grading ○ answering common questions ○ monitoring student engagement • allows instructors to focus on ○ mentoring ○ complex instruction ○ curriculum design
Limitations and Concerns	• reduced human interaction • heavy reliance on AI could ○ reduce student–teacher contact ○ weaken mentorship and emotional support • human interaction remains important for motivation and engagement • data privacy • AI systems require large amounts of student data • surveillance • misuse of personal information • data security

	• algorithmic bias • AI models may reproduce biases present in training data ○ unfair grading ○ unequal learning recommendations • over-automation ○ instructors may lose control over teaching decisions ○ students may become overly dependent on automated guidance
Future Directions for AI in Education	• human-AI collaboration ○ AI should assist instructors rather than replace them ▪ AI handles routine tasks ▪ humans handle complex teaching and mentoring • explainable AI ○ educational AI systems should be transparent ○ students and instructors should understand ▪ why recommendations are made ▪ how decisions are produced • ethical governance • AI in education requires ○ clear policies ○ ethical guidelines ○ responsible data use
Core Takeaway	• artificial intelligence can transform education by enabling ○ personalized learning ○ scalable instruction ○ automated support systems • AI should augment human teaching rather than replace • careful attention must be given to ethics, bias, and privacy